

Anirban Bhaduri

Office: LeConte LC413, UofSC
Columbia, South Carolina, U.S.A.

ABHADURI@email.sc.edu
anirbanbhaduri1905@gmail.com

Education

University of South carolina

Doctorate in Philosophy (Ph.D.)(Advisor: Dr.Matthew Ballard)

January 2021 – Present

Columbia, South Carolina,U.S.A.

University of Calcutta

Masters in Science (M.Sc.)

August 2017 – June 2019

Kolkata, West Bengal, India

University of Calcutta

Bachelor of Science (B.Sc.)

August,2014 – July,2017

Kolkata, West Bengal, India

Research Interests

My Research interest lies in the intersection of Derived categories, Algebraic Geometry, Representation Theory, Non-Commutative and Homological Algebra.

Publications and Preprints

- Bhaduri A., Dey S., Lank P. *Preservation for generation along the structure morphism of coherent algebras over a scheme*—**Bulletin of the London Mathematical Society** (2025)
- Bhaduri A., Goldberg I., Robotis A.A. *Dimensions Theory of Noncommutative Curves*—**submitted**.
- Bhaduri A., Davidov Y., Faber E., Honigs K., McDonald P., Overton-Walker C.E., Spence D. *An Explicit Derived McKay Correspondance For Some Complex Reflection Group of Rank Two*—**submitted**.
- Ballard M.,Bhaduri A. *Orlov Spectra for Weighted Projective Line*—**In Preperation**.

Other Experience

Graduate Teaching Assistant

January,2021 – December,2021

University of South Carolina

Graduate Instructional Assistant

January,2022 – Present

University of South Carolina

Graduate Peer Mentor

January,2023 – April,2024

University of South Carolina

Presentations and Talks

Graduate Colloquium, Spring 2025

Department of Mathematics, University of South Carolina

Georgia Algebraic Geometry Symposium(G.A.G.S), Spring 2025

Department of Mathematics, University of Georgia

Syzygies and Mirror Symmetry(S.M.S) Virtual Seminar, Spring 2025

Online Seminar

Derived Obsessed Graduate Student(D.O.G.S) seminar, Spring 2025
Online Seminar

Graz Algebra and Geometry Seminar, Fall 2024
Department of Mathematics, University of Graz

AGNES 2024, Fall 2024
Department of Mathematics, Dartmouth College

Graduate Colloquium, Spring 2023
Department of Mathematics, University of South Carolina

AGCA Seminar, Spring 2022
Department of Mathematics, University of South Carolina

Teaching

Instructor

University of South Carolina

- MATH 152 — Calculus Workshop
- MATH 170 — Finite Mathematics
- MATH 115 — Pre-Calculus Mathematics
- MATH 142 — Calculus II
- MATH 122 — Calculus for Business Administration
- MATH 111 — Basic College Mathematics
- MATH 141 — Calculus I
- MATH 115 — Pre-Calculus Mathematics

Columbia, South Carolina, U.S.A.

Fall 2024
Spring 2024
Fall 2023
Summer 2023
Spring 2023
Fall 2022
Summer 2022
Spring 2022

Teaching Assistant

University of South Carolina

- MATH 141 — Calculus I
- MATH 141 — Calculus I
- MATH 141 — Calculus I

Columbia, South Carolina, U.S.A.

Fall 2021
summer 2021
Spring 2021

Conferences and Workshops

Georgia Algebraic Geometry Symposium (G.A.G.S.) 2025 March 21-23, 2025. University of Georgia.

Algebraic Geometry Northeastern Series (A.G.N.E.S.) 2024 November 8-10, 2024. Dartmouth College.

Introductory Workshop: Noncommutative Algebraic Geometry February 5-9, 2024. Simons Laufer Mathematical Sciences Institute, formerly Mathematical Sciences Research Institute.

Connections Workshop: Noncommutative Algebraic Geometry February 1-2, 2024. Simons Laufer Mathematical Sciences Institute, formerly Mathematical Sciences Research Institute.

Mathematics Research Communities - 2023 June 4 -10, 2023. American Mathematical Society.

Concluding Conference of Simons Collaboration on Homological Mirror Symmetry April 25-28, 2023. Simons Center for Geometry and Physics, Stony Brook University.

Derived Categories, Moduli Spaces and Hyperkahler Varieties Minischool(July 30-31,2022) and Conference(August 1-5,2022). University of Michigan.

Homological Mirror Symmetry April 4-8, 2022. University of Miami, 5030 Brunson Drive, Cosford Cinema.

Analysis, Algebra, Number Theory and Probability Workshop February, 2019. Department of Mathematics, Rahara Ramkrishna Mission Vivekananda Centenary College, Kolkata, West Bengal, India.

Organization

Derived And (Non)Commutative Enthusiasts (D.A.N.C.E.)- Spring 2025, Online Seminar
Mathematical Research Communities Sectional at Joint Mathematics Meeting- Spring 2024
Algebraic Geometry Commutative Algebra and Number Theory Graduate Student Seminar- Fall 2023, University of South Carolina